COMPUTER STUDIES

INTRODUCTION TO COMPUTER (COMPUTER FUNDAMENTALS)

1.1 COMPUTER



Is an electronic device that processes, stores and produces data / information for the user. This is also referred as Central Processing Unit (CPU).

Inside the computer / CPU there is a Microprocessor Chip. The microprocessor chip is often called the "brain" of the computer as instructions from the software programs and input from the user are received and executed here.



What is the difference between Data and information?

Data:

Information:

1.2 USES OF COMPUTERS IN OUR DAILY LIFE

Computer Technology is used in a variety of equipment used in daily activities, such as:

1. Is used to store data for the user e.g. Files like music, documents, videos, pictures etc
2. Equipment to diagnose problems with automobile engines.
3. Automated Teller Machines (ATMs) to handle deposits, withdrawals, or bill payments.
4. Point-of-sale machines that process credit card and debit transactions in restaurants or stores.
5. Global Positioning Satellite (GPS) navigation tools in motor vehicles.
6. Robotics used by manufacturing firms to perform many tasks.
7. Medical equipment such as heart monitors and diagnostic scanners.
8. Small home appliances such as microwaves, coffee machines, and entertainment systems.
9. Large home appliances such as stoves/ovens, refrigerators, and washing machines.

1.3 TYPES OF COMPUTER

Many computers are designed specifically as computing devices, while others are embedded in products such as automobile engines, medical or industrial equipment, household appliances, or calculators. The main types are as follows;

1. DESKTOP COMPUTERS

Desktop computers are also called Personal Computers (PC) and sit on, beside, or under a desk.



Advantages of Desktop Computers

i. They are durable (not easily broken)
ii. They are fairly cheap to purchase
iii. It has prolonged performance rate though poor in speed

Disadvantages of Desktop Computers

iv. They too heavy to be carried around (not portable for remote use)
v. They need direct power source
vi. They have low graphic quality and poor speed interms of performance.

2. LAPTOP COMPUTERS

Laptop Computer is a computing device that fits within the lap of the user and is portable.



Advantages of Laptop Computers

i. They are portable(easy to carry)
ii. They are rechargeable (can be used in remote places where there is no electric power)
iii. They have high rate of speed and good graphic quality comparing to desktop computer.

Disadvantages of Laptop Computers

iv. They are easy to be stolen since are portable
v. They are easily broken (they are too fragile)
vi. They are abit costly comparing to desktop computers

3. PALMTOP COMPUTERS

Palmtop Computer is a computing device that fits within the palm of the user and is portable. Palmtop computers come with a number of pre-installed office applications and PDA.





Advantages of Laptop Computers

i. They are portable(easy to carry)
ii. They are rechargeable (can be used in remote places where there is no electric power)
iii. They have fairly rate of speed and good graphic quality comparing to laptop computers.

Disadvantages of Laptop Computers

iv. They are easy to be stolen since are portable
v. They are easily broken (they are too fragile)
vi. They are abit costly comparing to desktop computers

4. HANDHELD / MOBILE COMPUTERS




These are all electronic computing devices that Perform modern technological tasks in our daily life activities. These includes; cellular phones, tablets,

You may use a hand-held device as a telephone only, to send or receive voice or electronic messages, or as a portable personal computer. Many of these devices also offer the option to copy or download music or electronic books from the Internet.






1.4 COMPONENTS OF THE COMPUTER

These are all electronic devices that help the computer to work efficiently.

1. MONITOR (Output)

The monitor is an output device that enables you to view information the computer displays.



All monitors have a power switch as well as Brightness and contrast controls to adjust the screen Image

TYPES OF MONITORS

There are two types of monitors, namely;

i. **Flat screen Monitors** e.g. Plasmic Monitors, LCD Monitors



Plasimc Monitor





LCD Monitors

ii. Cathode Ray Tube (CRT) Monitors





CRT Monitors

2. **MOUSE (Input)**

Mouse is a pointing device enables the user to select or activate items on the screen. Moving the mouse on a flat surface such as a desk causes the mouse to initiate movement reflected by the pointer on the monitor.





Touchpad *is a* device that enables you to use your finger to move the mouse pointer around on the screen.

A *Mouse* and *touchpad* has two buttons that work in the same manner as the left and right buttons on a mouse.

Left Button:

- Is used to select and activate items in the computer or the windows desktop.
- Is used as Enter / Ok button

Right Button:

- Is used to display the shortcut menu for the item selected.

Middle Wheel / Scroll Wheel Button:

- Is used to scroll or move through the contents on the screen.
- Is used to zoom in and out images in some applications
- Is used to adjust volume in some media players application

3. **KEYBOARD (Input)**

Keyboard is an input device that allows the user to send information to the computer. It is the primary tool for inputting data.



The following provides a basic description of the more commonly used keys on the Keyboard;

1. **ESC** Cancels a current selection or generates a special code for the computer; also referred to as the *Escape* key.

2. **F1 –F12** *Function Keys;* These are located across the top of the keyboard and are labeled through. Each application program assigns a special meaning or function to each key, which is generally a shortcut for commonly used commands.

3. **Backspace** Deletes one character to the left of the cursor each time you press it.

   Delete Deletes the character above or to the right of the cursor each time it is pressed.

4. **Tab** Advances the cursor to the right by a set amount ($^1/_2$ inch) or to the next cell in a spreadsheet program.

   Also add up rows in table when pressed

5. **Ctr** *Control;* Provides a secondary meaning or function for almost every other key on the keyboard.

6. **Windows Key** Displays the Start menu.

7. **Alt** *Alternate;* Provides an alternate meaning or function for almost every other key on the keyboard.

8. **Spacebar** Inserts a blank space between words (per character), and is the safest key to press when a software program prompts you to press any key.

9. **Enter** Executes a typed command or a selected option from a Menu (Is an OK button).

   Marks the end of a line

   Creates a blank line between lines.

10. **Cursor Movement Keys** Moves the cursor up, down, left, or right, and is usually located between the typewriter keys and the numeric keypad.

11. **Numeric Keys** This is located at the far right of the keyboard, and can be toggled on and off by pressing the key in the pad.

12. **Caps Lock** Locks the letter keys to produce only uppercase versions of each letter.

13. **Shift Key** Displays uppercase letters for the alphabetic keys or the punctuation symbols that share the number keys.

4. **SPEAKER (Output)**

Speakers play the sounds saved as digital files on the computer. Different forms of sound files, such as mp3, wav, or wma, can be specific to music programs or generic for any player device on the computer.



5. **PRINTER (Output)**

Printer is an electronic device that converts what is on the screen / in the computer into print when you activate the print command.



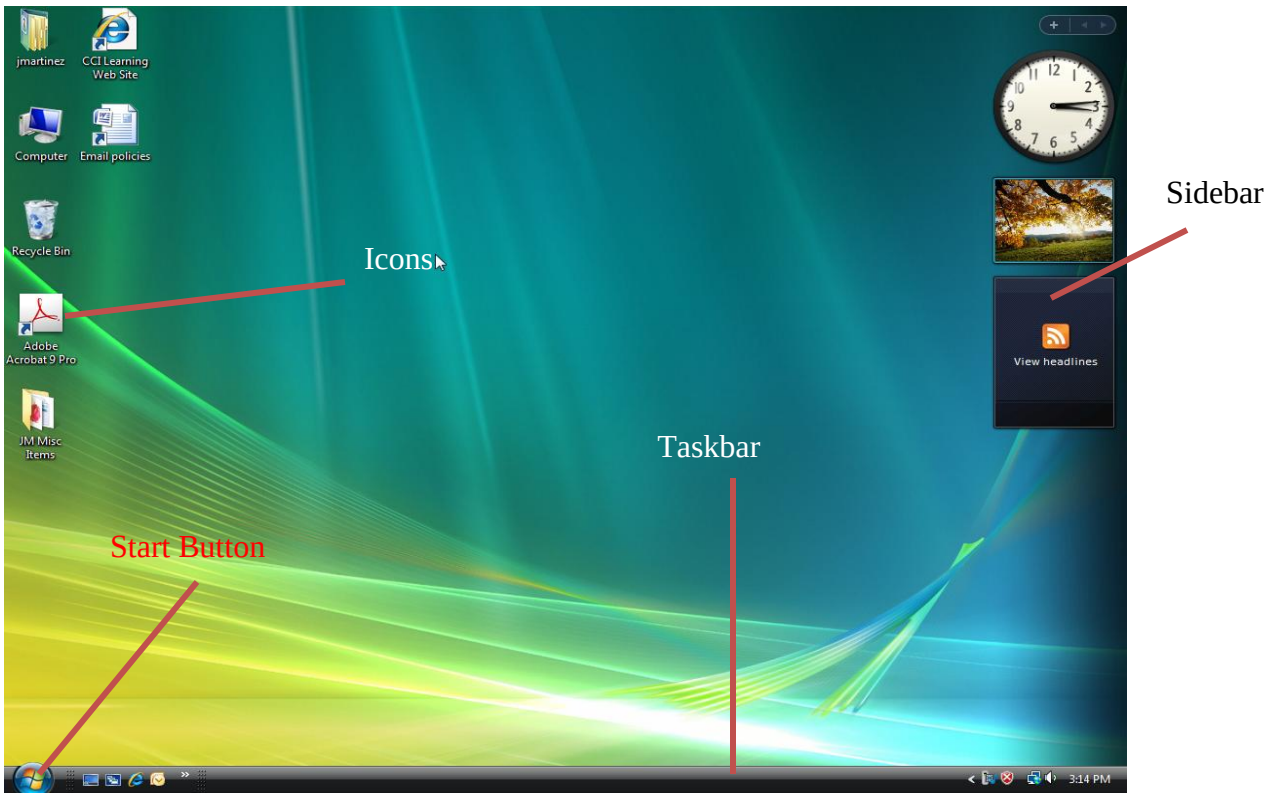
6. **SCANNER (Input)**

Scanner is an electronic device that takes a picture of the original item and converts it to a digital format.



1.5 EXPLORING THE WINDOWS DESKTOP

Windows Desktop is the working area or screen on which all works are done. Windows Desktop consist of, icons, taskbar and the sidebar.



Desktop Icons

These are "shortcuts" to frequently opened / used programs, folders, or files in the computer.

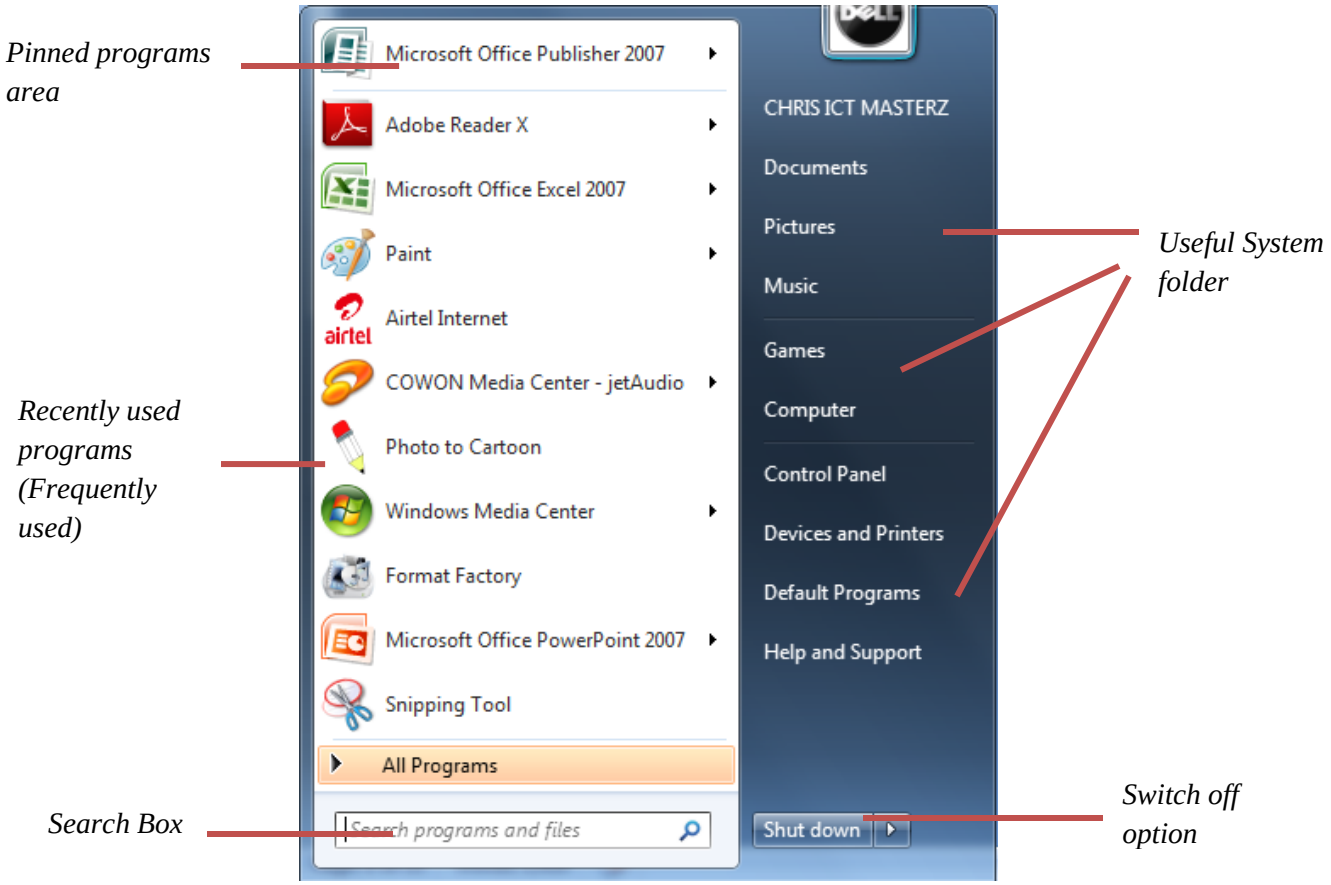
Windows Sidebar

This area displays mini-programs or gadgets, such as a clock, a calendar, RSS feeds, weather updates, and so on.

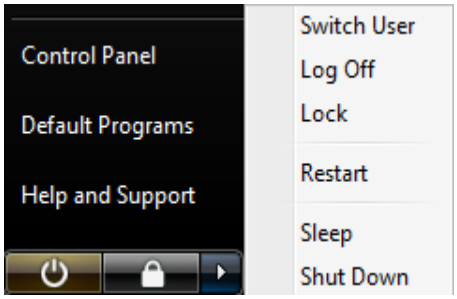
Taskbar

The taskbar is where the Start button, taskbar notification area, and sometimes Quick Launch toolbar or other toolbars reside.

1. **Start Button**
   This is where all compute programs are found and used to start programs, open documents, find items on your computer, and get help, as well as log off and shut down your computer. You can use the mouse or the keyboard to navigate through the Start menu.



Exiting the Computer Properly;



From a security perspective, it is extremely important to save your files, close the programs, and log off the computer when you finish working to prevent unauthorized access to your files and, more importantly, to your company's network.

Shut Down Closes all open items and shuts down the computer.

Restart Closes all open items and restarts the computer; also called a reboot or warm boot.

Hibernate Available only on notebooks, this saves what is on the desktop and then shuts down the computer completely so it consumes no power. When you restart the computer and log on, the desktop returns to where you left it.

Sleep Puts the computer in a state where it consumes less power without losing your place on the screen; the computer does not shut down in this mode. If you are using a desktop computer, the Sleep command appears as Standby.

Switch User Switches to another user account without logging out of the current account.

Log Off Closes all open items, logs out of the current user account, and returns to the log on screen.

Lock Hides the desktop behind a log on screen.

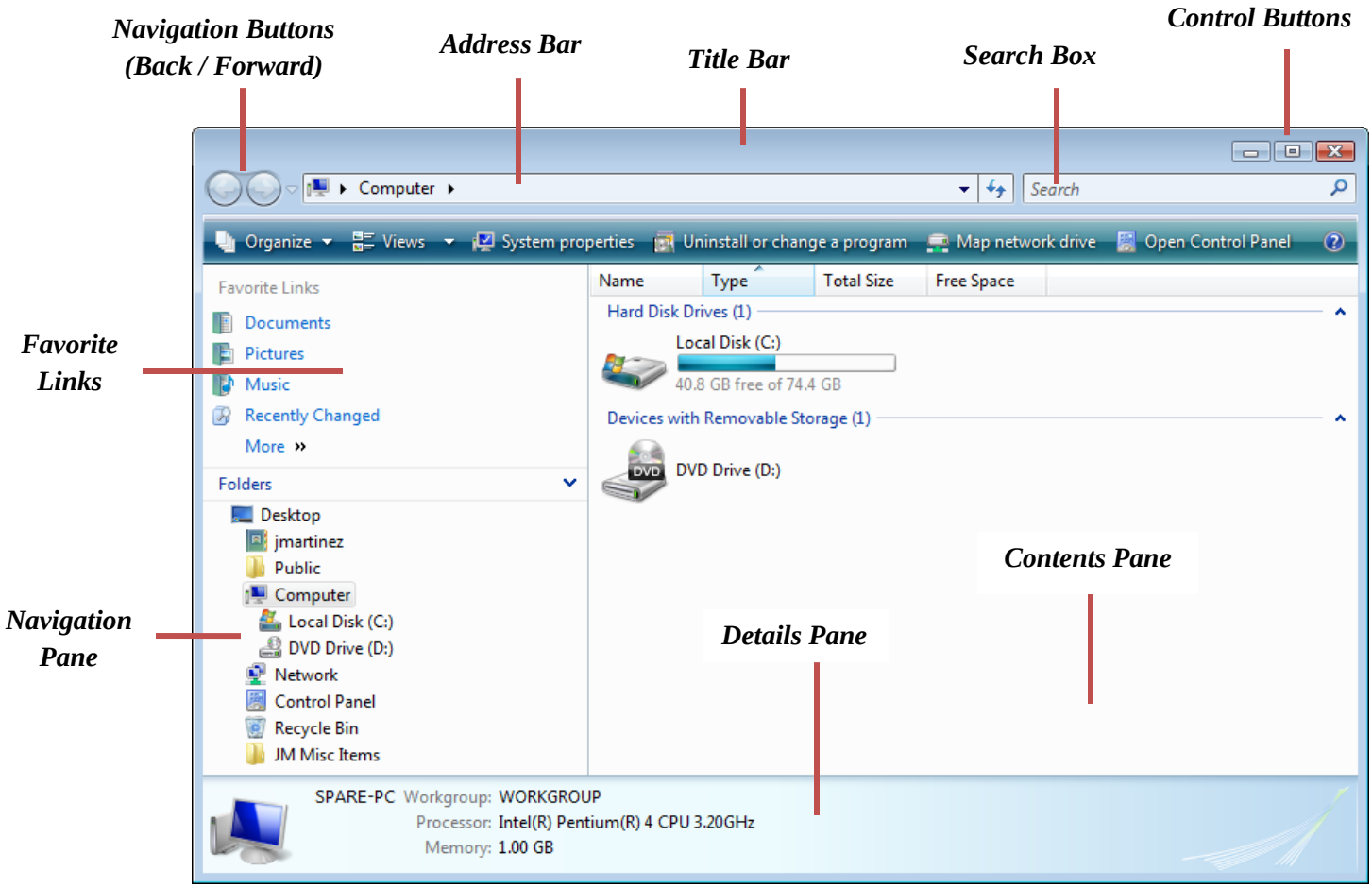
2. **Notification area**
   Indicate program status, or provide another shortcut to already opened programs small icons, sometimes called the *System Tray*.

Quick Launch Toolbar

This is another area for shortcuts that launch programs, open documents, or open folders.

Looking at a Typical Window;

When programs or folders are opened, they appear on the desktop in individual "windows." You can have multiple windows displayed on the screen, but each window will share similar features.



Navigation Buttons (Back and Forward Buttons)

Allow you to navigate back or forward to display different views of files and folders.

Title Bar

Displays the name of the currently active feature or application program (for example, Computer).

Command Bar

Provides commands you can use to organize, view, or protect your data.

Navigation Pane

Displays folders and drives that you can double-click in order to see their contents.

Contents Pane

Displays the contents of the folder or drive selected in the Navigation Pane.

Details Pane

Displays properties or details about the selected file or folder in the Contents pane.

Control Buttons

Change the way currently open windows are displayed, as follows;

 (Minimize)	Temporarily closes the window, replacing it as a button on the taskbar.
 (Maximize)	Displays the window in full screen.
 (Restore Down)	Restores the window to the size it was before it was maximized.
 (Close)	Closes the window.

Creating Folders;

You can create folders at any level, including directly on the Windows Desktop. To create a folder, use one of the following methods;

- On the Command bar, click **Organise** and then click **New Folder**, or
- right-click the location (i.e., drive or folder) in the Folders list for the new folder, and then click **New**, **Folder**, or
- right-click a blank area of the Contents pane and then click **New**, **Folder**.

Renaming a Folder;

The name of a folder can be as long or short as required (up to 255 characters), although with longer names, the entire title may not be visible in certain views.

To change a folder's name, use one of the following methods:



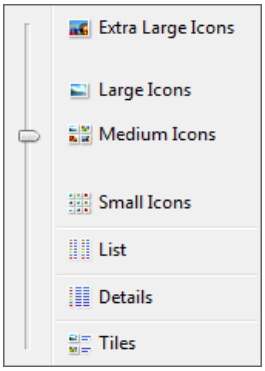
- Click the folder icon to select it and then press to activate the Edit mode, or
- click the folder icon, then click inside the folder name to activate the Edit mode, or
- right-click the folder and then click **Rename** from the shortcut menu.

Changing the view of a file or folder;

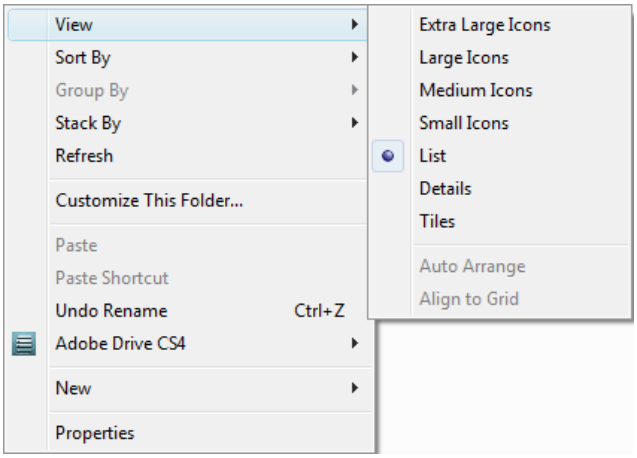
There are different ways to display information for files or folders. Occasionally you may want to sort files in a specific order, or to see more information for the files or folders. Select the view options using one of the following methods;

- Click the arrow 

of the button on the Command bar, or



- right-click in a blank area of the Contents pane and then **View**, or



- on the Command bar, click the **Views** button to cycle through the different views.
  i. Extra Large Icons
  ii. Large Icons
  iii. Medium Icons

1.6 FILE TRANSFERING

This is the process of moving files from different locations in the computer or from the removable storages to other specified locations, either in the computer or removable storages. The following are main commonly used ways of transfering files;

i. **Copy** + Paste Duplicating files to another location.
   - Click 

and then **Copy**, navigate to the new location and then click 

, select **Paste**, or
   - Press **Ctrl** + **C** , move to the new location and then press **Ctrl** + **P** , or
   - Right-click the selection and then click **Copy**, navigate to the new location, right-click and then click **Paste**.

ii. **Cut** + Paste Moving the files completely to the specified location.

- Click  and then **Cut**, navigate to the new location and then click  , select **Paste**, or
- Press **Ctrl** + **X**, move to the new location and then press **Ctrl** + **P** , or
- Right-click the selection and then click **Cut**, navigate to the new location, right-click and then click **Paste**.

iii. **Drag Method** transfering files with an effort by long pressing the left button.

- Select the item and long press with the left button on the item and move it to the new location and release your finger in the new location.

iv. Send-to- An option from the option menu that directs the selected file to the desired location.

- Right press the item and then click **send-to** option, then a sub-menu pops where you will seletect the removable device attached to the computer.

1.7 OPERATING SYSTEMS

An *operating system* is a collection of programs designed to control the computer's interaction and communication with the user. Without Operating System a computer is useless. Modern operating Systems use a **graphical user interface** or **GUI** (*Pronounced gooey)*. A GUI lets you use your mouseto click icons, buttons and menus, and everything is clearly dispalyed on the screen using a combination of graphics and text.

Types of Operating Systems

The Three (3) most common Operating systems are;

1. **Linux OS**
   Linux (*Pronounced* LINN-ux) is a family of open-source operating systems which means they can be modified and distributed by anyone around the world. The advantage is that it is free and there are many different distributions.

2. **Microsodt Windows OS**

Microsoft created Windows Operating System in the mid – 1980s. There have many different vrsions of Windows, mostly;

(i) Windows XP (1980s)
(ii) Windows Vista (2007)
(iii) Windows 7 (2009)
(iv) Windows 8 (2012)
(v) Windows 10 (2015)



Windows Vista

Windows Operating System comes pre-loaded on most new PCs, which helps to make it the most popular Operating System in the world.

3. **Mac OS**

Mac Operating System which previously was called **OS X**, is a line of Operating systems created by Apple. It comes pre-loaded on all Mciintosh Computers , or Macs. Some of its specific versions include;

(i) Sierra (2016)
(ii) High Sierra (2017)
(iii) Mojave (2018)


Finder File Edit View Go Window Help
Mon 5:12 PM
Macintosh HD
Applications
DEVICES
Macintosh HD
iDisk
PLACES
Applications
Desktop
guest1
Documents
SEARCH FOR
Today
Yesterday
Past Week
All Images
All Movies
All Documents
Address Book
Application
Name
Date Modified
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Kind
Address Book 14-Feb-08, 6:49 AM 47.6 MB
Adobe Acrobat 8 Professional 12-May-08, 6:35 PM --
Adobe Bridge CS3 15-May-08, 11:42 PM --
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